🏫 ASP.NET Core MVC – Basic Assignments

🔹 1. Create Controller and Action Methods

✅ Assignment 1: Basic Controller

Scenario: Build a simple website.

Tasks:

1. Create a controller named HomeController
2. Add action methods:
    - Index() → return "Welcome to ASP.NET Core MVC"
    - About() → return "This is About Page"
    - Contact() → return "Contact us at support@test.com"

✅ Assignment 2: Multiple Controllers

Scenario: Education system

Tasks:

1. Create:
    - StudentController
    - TeacherController
2. Add:
    - StudentController → Index(), Profile()
    - TeacherController → Index(), Details()

👉 Call each action using browser URL.

🔹 2. Create Action Methods with Parameters

✅ Assignment 3: Passing Simple Parameters

Scenario: Display user details

Tasks:

1. Create UserController
2. Add action:
3. public IActionResult Details(string name, int age)
4. Display:
    - Name
    - Age

👉 Example URL:

/User/Details?name=John&age=25

✅ Assignment 4: Route Parameters

Scenario: Product details page

Tasks:

1. Create ProductController
2. Add action:
3. public IActionResult GetProduct(int id)
4. Return:
    - "Product Id is: {id}"

👉 Try:

/Product/GetProduct/101

✅ Assignment 5: Multiple Parameters

Scenario: Calculator

Tasks:

1. Create MathController
2. Add actions:
    - Add(int a, int b)
    - Multiply(int a, int b)
3. Return result

🔹 **3. Create View**

✅ **Assignment 6: Basic View**

Scenario: Show welcome page

Tasks:

1. Create HomeController
2. Create view for Index
3. Display:
4. <h1>Welcome to MVC Application</h1>

✅ **Assignment 7: Passing Data to View**

Scenario: Student page

Tasks:

1. Create StudentController
2. Action:
3. public IActionResult Details()
4. Pass:
    - Name
    - Age
5. Display in view

🔹 4. Use ViewData Object

✅ Assignment 8: ViewData Basics

Scenario: Show user info

Tasks:

1. In controller:
2. ViewData["Name"] = "John";
3. ViewData["Age"] = 25;
4. Display in View using:
5. @ViewData["Name"]

✅ Assignment 9: Dynamic Page Title

Scenario: Dynamic titles

Tasks:

1. Set:
2. ViewData["Title"] = "Home Page";
3. Display in layout or view:
4. <title>@ViewData["Title"]</title>

✅ Assignment 10: Passing Multiple Values

Scenario: Employee details

Tasks:

1. Pass:
    - Name
    - Salary
    - Department
2. Display all using ViewData

🔹 **5. Razor Page (Razor Syntax)**

✅ **Assignment 11: Razor Syntax Basics**

Scenario: Display current date

Tasks:

1. In View:
2. Today: @DateTime.Now

✅ **Assignment 12: Conditional Rendering**

Scenario: Check age eligibility

Tasks:

1. Pass age from controller
2. In view:
3. @if(age >= 18)
4. {
5. Eligible to vote
6. }
7. else
8. {
9. Not eligible
10. }

✅ **Assignment 13: Loop in Razor**

Scenario: Display list of students

Tasks:

1. Pass list from controller

2. Use:
3. @foreach(var student in students)
4. {
5. @student
6. }

✅ Assignment 14: Strongly Typed View (Intro)

Scenario: Show student object

Tasks:

1. Create model:
2. public class Student
3. {
4. public string Name { get; set; }
5. public int Age { get; set; }
6. }
7. Pass model to view
8. Use:
9. @model Student

🎯 Mini Project (Combine All Concepts)

✅ Assignment 15: Student Management Page

Tasks:

1. Create StudentController
2. Action:
3. public IActionResult Details(string name, int age)
4. Use:

- ViewData for additional info
- Razor for condition & display

5. Create View to show:
    - Name
    - Age
    - Eligibility (age > 18)